Check for updates

ASME Journal of Biomechanical Engineering
Online journal at:
https://asmedigitalcollection.asme.org/biomechanical

Thor E. Andreassen
Center for Orthopaedic Biomechanics,
Department of Mechanical and
Materials Engineering,
University of Denver,
Denver, CO 80210
e-mail: thor.andreassen@du.edu

Peter J. Laz
Center for Orthopaedic Biomechanics,
Department of Mechanical and
Materials Engineering,
University of Denver,
Denver, CO 80210

Ahmet Erdemir
Department of Biomedical Engineering and
Computational Biomodeling (CoBi) Core,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Thor F. Besier
Department of Engineering Science,
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand

Jason P. Halloran
Applied Sciences Laboratory,
Institute for Shock Physics,
Washington State University,
Spokane, WA 99164

Carl W. Imhauser
Department of Biomechanics,
Hospital for Special Surgery,
New York, NY 10021

Snehal Chokhandre
Department of Biomedical Engineering and
Computational Biomodeling (CoBi) Core,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Ariel Schwartz
Department of Biomedical Engineering and
Computational Biomodeling (CoBi) Core,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Deciphering the "Art" in Modeling and Simulation of the Knee Joint: Assessing Model Calibration Workflows and Outcomes

Model reproducibility is a point of emphasis for the National Institutes of Health (NIH) and in science, broadly. As the use of computational modeling in biomechanics and orthopedics grows, so does the need to assess the reproducibility of modeling workflows and simulation predictions. The long-term goal of the KneeHub project is to understand the influence of potentially subjective decisions, thus the modeler's "art", on the reproducibility and predictive uncertainty of computational knee joint models. In this paper, we report on the model calibration phase of this project, during which five teams calibrated computational knee joint models of the same specimens from the same specimen-specific joint mechanics dataset. We investigated model calibration approaches and decisions, and compared calibration workflows and model outcomes among the teams. The selection of the calibration targets used in the calibration workflow differed greatly between the teams and was influenced by modeling decisions related to the representation of structures, and considerations for computational cost and implementation of optimization. While calibration improved model performance, differences in the postcalibration ligament properties and predicted kinematics were quantified and discussed in the context of modeling decisions. Even for teams with demonstrated expertise, model calibration is difficult to foresee and plan in detail, and the results of this study underscore the importance of identification and standardization of best practices for data sharing and calibration.
[DOI: 10.1115/1.4063627]

Keywords: computational modeling, reproducibility, knee biomechanics, finite element analysis, joint mechanics, tissue mechanics, ligament

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

[1]Corresponding author.
Manuscript received April 17, 2023; final manuscript received August 7, 2023; published online October 26, 2023. Assoc. Editor: Spencer P. Lake.

 Copyright © 2023 by ASME

Neda Abdollahi Nohouji
Department of Biomedical Engineering and
Computational Biomodeling (CoBi) Core,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Nynke B. Rooks
Department of Engineering Science,
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand

Marco T. Y. Schneider
Department of Engineering Science,
Auckland Bioengineering Institute,
University of Auckland,
Auckland 1010, New Zealand

Shady Elmasry
Department of Biomechanics,
Hospital for Special Surgery,
New York, NY 10021

William Zaylor
Department of Biomedical Engineering,
Lerner Research Institute,
Cleveland Clinic,
Cleveland, OH 44195

Donald R. Hume
Center for Orthopaedic Biomechanics,
Department of Mechanical and Materials
Engineering,
University of Denver,
Denver, CO 80210

Kevin B. Shelburne[1]
Center for Orthopaedic Biomechanics,
Department of Mechanical and Materials
Engineering,
University of Denver,
Denver, CO 80210
e-mail: kevin.shelburne@du.edu

1 Introduction

Addressing the inability to reproduce scientific findings has been a key focus of the National Institutes of Health (NIH) over the last decade [1]. As the use of computational modeling in biomechanics and orthopedics grows, so does the need to assess the variations in modeling and simulation workflows and reproducibility of model predictions. Knee models consist of anatomical structures typically digitized and represented by meshes, whose material behavior is formulated by constitutive equations. A use case of interest is simulated by subjecting the model to prescribed kinematic and kinetic conditions and observing the response. Models provide insight into the form-function relationships in the knee, enabling predictions of patient-specific knee mechanics in health, injury, and disease. Nonetheless, variations in modeling decisions and reproducibility are concerns because computational models are applied in a range of applications from scientific exploration to clinical tools, such as the development of treatment strategies, the design and evaluation of orthopedic implants, and support for surgical planning. Even so, publications addressing reproducibility are uncommon [2–4], and particularly rare in human simulation [5–7].

With the need for improved reproducibility in modeling and simulation, the KneeHub project is investigating the "art" in modeling and simulation workflows to understand the influence of the approaches and decisions of the modelers on the computational models and their consequent outputs [5]. Five independent teams worked with the same data to build and test computational knee joint models with prescribed simulation goals in four phases: *model development*, *model calibration, model benchmarking*, and *model reuse* (Fig. 1). The model development phase, described in Rooks et al. [6], aimed to deliver an initial working model to simulate passive knee flexion of two cadaver specimens (Fig. 2). The model *calibration phase*, which is described in this article, aimed to calibrate the initial models to experimental data obtained from joint mechanics tests of the modeled specimens. The model benchmarking phase evaluates model performance against experimental data obtained from the specimens, which were intentionally separated from data used in previous phases. Finally, in the model reuse phase, teams will use their models for in-silico investigations that have scientific and clinical relevance. Throughout all phases of the project, the teams prospectively document their decisions, including

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

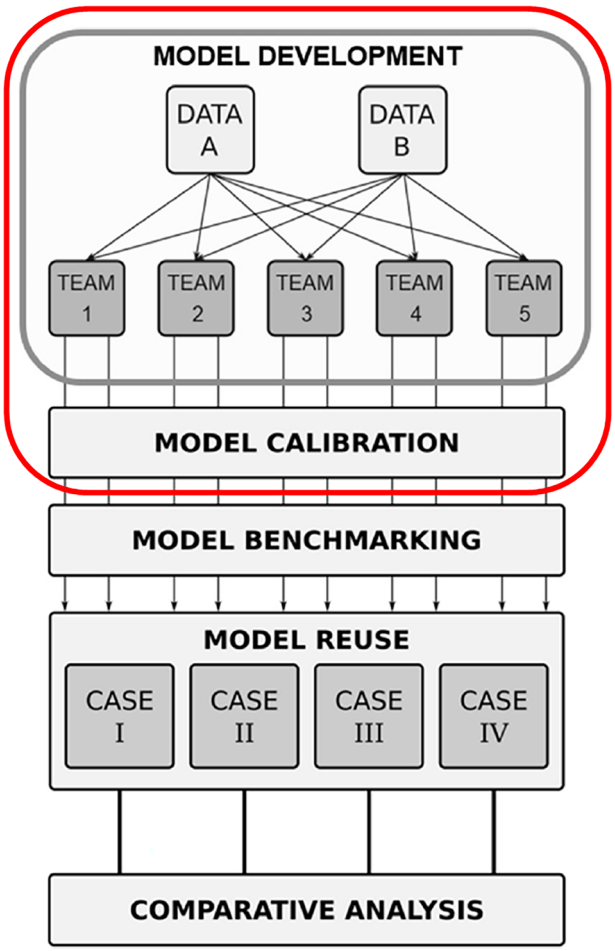

Fig. 1 Phases of the KneeHub project. Adapted from Rooks et al. [6]. DATA A refers to Natural Knee specimen DU02 and DATA B refers to Open Knee(s) specimen OKS003.

deviations from the proposed plans, and publicly share their models and simulations[2].

Calibration commonly involves the application of known kinematic-kinetic conditions to a model in a computer simulation, and the tuning of parameters in the model to achieve a measured outcome, with our particular focus on reproducing the complex dynamic behavior of the human knee. Computational models of how the knee responds to applied loads require calibration to experimental load–displacement data to determine the parameters of elastic soft tissue representations of the knee ligaments and cartilage. For example, ligaments are often represented by one-dimensional (1D) point-to-point springs with uncertain stiffness, slack length or reference strain, and attachment sites [8–15]. Ligaments may also be represented by three-dimensional (3D) structures with uncertain prestretch and differing material constants [6,16]. Direct measurement of the mechanical parameters that dictate the nonlinear behavior of the ligaments is often difficult or impossible. Instead, a common method to calibrate these uncertain parameters is to match the behavior of the knee model with experimental measurements of laxity or the nonlinear force–displacement characteristics of the knee under various loads and knee flexion angles [17]. Laxity is measured as the translations and rotations of the tibiofemoral joint under a known load applied along a single degree-of-freedom (DOF) [18–23]. During calibration, ligament properties are adjusted in the model until the tibiofemoral motion matches the measured response. Frequently, the measured force–displacement curves for multiple DOFs at different flexion/extension angles are resampled to create target data points for use in an optimization. A cost function is created to minimize the difference between the simulated and measured specimen-specific tibiofemoral displacement(s) and/or load(s) [8,24]. Model calibration choices are diverse, leading to challenges in reproducibility; for example, decisions are made related to the constitutive models of the tissues, which tissues and parameters are chosen for tuning, the initial values of these parameters, the choice of calibration targets, and the optimization methods.

This study reports the findings for the model calibration phase of the KneeHub project with three objectives. First, we compare the calibration process of the teams through a comparison of their model calibration workflows to understand the decisions that modelers made to deliver a model tuned to experimental data. Second, we compare the calibration targets chosen by each team from the experimental data. Third, we compare the calibrated ligament parameters and kinematics of the resulting models among all of the teams, to assess the impact of differing calibration decisions and processes.

2 Materials and Methods

2.1 Summary of the KneeHub Model Calibration Phase. The goal of the model calibration phase was to refine specimen-specific models of the knee by tuning parameters with documented workflows so that the model responses reproduced the observed experimental data. The teams started with their initial working models (the outcome of the previous model development phase), where the anatomy was defined from imaging data [6]. In the current

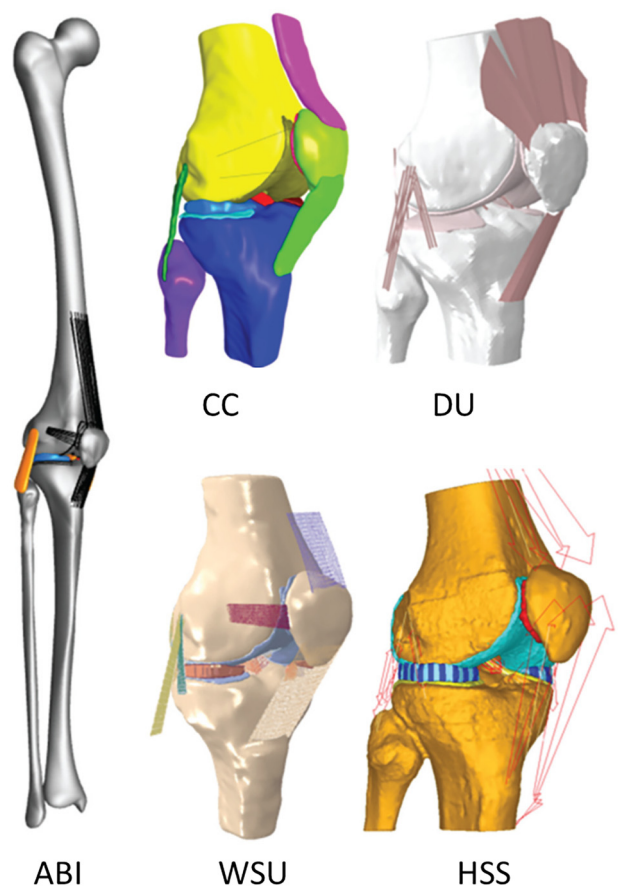

Fig. 2 Models built by each team at the conclusion of the model development phase of the KneeHub project. Adapted from Rooks et al. [6].

[2]https://simtk.org/projects/kneehub

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

phase, specimen-specific joint kinematics-kinetics data available in Open Knee(s) [25] and Natural Knee Data [24] were used to calibrate model parameters that may otherwise be missing or uncertain, e.g., properties related to ligament stiffness and reference strains. The calibration process inherently includes the challenge of interpreting and postprocessing the experimental specimen-specific measurements. The teams also chose to consider the available literature describing joint and tissue level function of the knee when selecting model parameter values and bounds. Each team was free to choose their own approach and independently develop their workflow, with sharing only after all teams completed their work. To facilitate comparison, the tibial and femoral coordinate systems for each team were made consistent with the convention of each experiment. The outcome of the calibration phase was two calibrated specimen-specific models from each team corresponding to two different specimen-specific data sources. Additional outputs included the set of updated model parameters, loading and boundary conditions used in the calibration-related simulations, as well as any changes in anatomical or tissue mechanical representations from the prior model development phase.

2.2 Knee Specimen Experimental Datasets. Each team developed and calibrated two knee models using existing data sets of medical imaging and mechanics—one model from Open Knee(s) [25,26] and one from Natural Knee Data [27]. In the model development phase of the project, the teams decided to use data from specimen OKS003 of Open Knee(s) and specimen DU02 of Natural Knee Data. Specimen OKS003 represents a left knee of a 25-year-old female donor (height: 1.73 m; mass: 68 kg; BMI: 22.8). Specimen DU02 represents a right knee of a 44-year-old male donor (height: 1.83 m; mass: 70.31 kg; BMI: 21.02). Open Knee(s) data include anatomical imaging using magnetic resonance imaging [28] and mechanical testing data of tibiofemoral joint kinematics-kinetics [29]: passive flexion (0 deg–90 deg), joint laxity (at 0 deg, 30 deg, 60 deg, and 90 deg flexion for internal-external (IE) and varus-valgus (VrVl) rotation and anterior-posterior (AP) translation), and combined loading (at 0 deg, 30 deg, 60 deg, and 90 deg flexion; permutations of IE and VrVl moments, and AP forces). Digitization was performed during mechanical testing that provided the coordinates of anatomical landmarks and point clouds on spherical registration markers. Natural Knee Data include computed tomography and T2-weighted magnetic resonance imaging. Experimental data consisted of digitized landmarks on the bone, cartilage, and ligament attachments and tibiofemoral joint kinematics-kinetics [24]: passive flexion (0 deg–120 deg), joint laxity at 0 deg, 15 deg, 30 deg, 45 deg, 60 deg, 75 deg, 90 deg, and 120 deg flexion for IE and VrVl rotation and AP translation, repeated passive flexion, and laxity testing after sequential resection of the cruciate ligaments. For both knees, however, only the joint laxity measurements for IE and VrVl rotation and AP translation of the intact knees were allowed to be used in the model calibration phase; the combined loading trials and those with sequential resection of ligaments were reserved for comparison during the model benchmarking phase. From the available measurements, model calibration was performed with kinematics-kinetics data selected by each team.

2.3 Model Calibration Phase Outputs. To improve model reproducibility, for each knee model, a data package was provided by each team along with documentation of any deviations to their a priori-written calibration specifications [5,26,30]. The data packages of each team contained the initial and final outcomes of the model calibration phase. The initial outcomes included working models, consisting of representations of anatomy (geometries of the modeled tissues and meshes), tissue behavior (e.g., constitutive models and tissue responses), interactions between model components (connectivity representations and contact models), anatomical coordinate systems, and loading and boundary conditions. The final outcomes included the working calibrated models and simulation results for team-chosen calibration targets and passive knee flexion from 0 deg to 90 deg; the latter facilitated comparison to the simulation results from the model development phase. The calibration output packages produced by each team, including documentation, calibrated models, any developed code, and simulation outputs, are publicly available on the KneeHub project SimTK page.

Teams prepared proposed specifications and protocol deviation documents capturing the model calibration phase. Minimal guidance was provided for these documents to allow teams to document as they saw fit, with the proviso that sufficient detail be provided for the fellow modelers to be able to reproduce the modeling process and trace the associated model decisions. The specifications document aimed to describe the steps taken to obtain a working specimen-specific model calibrated to the load–displacement behavior of the specimen from the data available. The specification document was prepared prior to performing the calibration process. The protocol deviations document recorded any deviations from the specifications that occurred during the model calibration process. The document included sufficient detail to describe and justify the protocol deviations, including the impact of the deviation on outputs and when the deviation was made.

2.4 Methods for Comparison. In prior work, Rooks et al. described the model development phase; differences in model generation were documented and variations in geometric representations of bones, cartilage, ligaments, and passive flexion kinematics were quantified [6]. In a second study by Rooks et al. [31], we compared the gross properties of the meshes, e.g., density and element types and material models used. In this work, comparisons were made between the documented model calibration workflows provided by each team and the final calibrated performance of the working models. Data from the specification and protocol deviations for each team were used to compare differences in the calibration strategy. Each team individually chose calibration targets derived from the experimental data; for comparison, the range of approaches was visualized in plots of the chosen targets for AP, IE, and VrVl laxity.

After calibration, the reference strain parameters for the cruciate and collateral ligaments consisting of the ACL, PCL, MCL, and LCL, were compared. The number of bundles and fibers used to represent each ligament differed between teams. For example, for the MCL, Team ABI and Team CC had one reference strain, Team HSS had two, Team DU had three, and Team WSU had four. To compare reference strains considering these differences, average, maximum, and minimum values were considered for each of the four ligaments.

Each team predicted passive flexion kinematics for each model and results were compared to the experimental measurements. The kinematic outputs of each team were transformed to the same coordinate system as the original experimental data and resampled in one-degree increments. The DU02 experimental coordinate system is absolute clinical Grood and Suntay joint coordinate system (GS) rotations and translations [32]. In contrast, the OKS003 experimental coordinate system is cylindrical GS joint rotations and translations relative to a neutral full-extension experimental pose [32]. Predicted AP, IE, and VrVl versus knee flexion angle of each team for both models were compared against the experimental data. Root mean squared error (RMSE) was calculated for each predicted kinematic output relative to the experimental data. Additional RMSE comparisons were made between the predicted AP, IE, and VrVl knee kinematics and the results for the uncalibrated models reported in prior work [6].

3 Results

The KneeHub project website provides an entry point for all activities related to the comprehensive study [26]. The project

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

workflow is organized there and includes source data and each individual team's documentation and models.

3.1 Differences in Calibration Strategies. The calibration process for all teams was designed to match the predicted displacement of the model at a given load and knee flexion angle with the experimental measurement at that configuration. To predict displacement, all teams applied the knee flexion angle in displacement control and applied all other DOFs in force control. Parameters defining ligament behavior were adjusted until the predicted displacement matched displacement targets chosen by each team from the experimental data.

3.1.1 Differences in Optimization Strategy. While all teams relied on optimization to perform calibration, not all groups used the same optimization strategy. Teams ABI, CC, HSS, and WSU utilized local or gradient-based optimization routines. Team DU used a local solver with restarts for the DU02 model and a global or metaheuristic optimization method (e.g., simulated annealing) for the OKS003 model. Team ABI was the only team that utilized multiple types of optimization methods for the same model. In total, 8 different types of optimization algorithms were used (Table 1). The optimization cost function was different for each team. While four teams (ABI, CC, DU, and WSU) used RMSE between experimental and model displacements for the cost function, one team (HSS) utilized the sum of 4th order power of each error. Team DU and Team WSU included an additional penalty constraint to the cost function in the form of a squared error term to ensure some load was observed in every ligament across the trials. Team ABI added penalties based on optimization convergence speed and model failure (Table 1). Stopping criteria were also different among the teams. Most teams utilized stopping criteria based on a function of the design variables, the RMSE, or both, while one team (ABI) chose the optimal solution within a set amount of computational time (Table 1).

3.1.2 Differences in Chosen Calibration Targets. The teams chose different sets of calibration targets based on their experience and desired outcomes of the model calibration for emphasis of certain features. For example, Team DU picked the maximum and minimum loads for a given laxity test across a range of flexion angles (Fig. 3). Teams HSS and WSU also used target points at multiple loads and flexion angles for various laxity trials. In contrast, Team CC calibrated to multiple loads at a single knee flexion angle near the full extension pose. Team WSU was the only team that split the chosen calibration points into separate sets used to calibrate the model and to validate the accuracy of the model to unseen trials. Additionally, they used an a priori analysis considering the ligament lengths to determine the specific points used for calibration or validation separately for each knee, yielding differences between the targets chosen for each knee. Team ABI was the only team to not directly use the experimental calibration points as targets and instead resampled data to create three-dimensional representations of knee flexion, load, and resulting displacement for each of the AP, IE, and VrVl laxity trial sets. They then fit a surface to the data, and target points were chosen as a range of loads at specific knee flexion angles for different types of laxity trials. The number of calibration targets used by the teams ranged between 8 and 93, with a mean of 34.4. Notably, the teams that used 3D ligament representations had more calibration targets than all teams using 1D ligament representations. Two teams calibrated to more than one DOF for a given laxity measurement; Team DU chose a secondary target DOF for each trial and Team WSU used targets from all 6 DOF. For example, Team DU used a cost function for AP translation that also included IE rotation. In contrast, Team HSS chose to focus on the primary loads for the specific DOF being simulated. For example, they applied 20 N of anterior force during the anterior laxity simulation with all other loads except joint compression held at zero (Fig. 3).

3.1.3 Differences in Calibrated Ligament Representations and Properties. For all teams, the calibration design variables consisted of parameters affecting ligament behavior. All teams calibrated the reference strains of the ligaments. Most teams used representations of stiffness or modulus from the literature. Only one team (DU) calibrated stiffness corresponding to the linear-elastic region of the ligament force–displacement curve in addition to the reference strains. No teams included the length or attachment sites of the ligaments as design variables in their optimization, instead opting for locations derived from the imaging and segmentation process. The number of design variables for the model optimization of each team ranged between 4 and 26, with a mean of 12.8 (Table 1). Notably, the teams that used 3D ligament representations had fewer design variables than all teams using 1D ligament representations. Teams CC and HSS chose specific calibration targets for the calibration of specific ligaments. For example, Team CC's anterior laxity targets at 0 degrees of knee flexion were used to calibrate the ACL reference strain, while leaving other ligament parameters constant. For the teams with 1D ligament representations, differences in the assignment of design variables to the fibers of different ligament bundles were observed. Team WSU calibrated the parameters of each fiber of a ligament bundle separately. In contrast, Team DU and Team HSS calibrated the parameters as a single set for all fibers of a given bundle. Team CC was the only group that performed unconstrained optimization of the ligament parameters, while the other teams constrained the optimization with a set of upper and lower bounds for each ligament parameter.

3.2 Comparison of Calibration Results

3.2.1 Ligament Stiffness and Reference Strain. A broad range of ligament stiffness values was applied in the models (Table 2). Only one team (DU) chose to include ligament stiffness in the design variables of the calibration. Even so, for example, the value of stiffness in the anterior-medial fibers of the ACL of Team DU was approximately twice that of Team HSS. Likewise, for the two teams using 3D continuum representations, the ACL modulus applied by Team CC was roughly twice that of Team ABI (Table 2).

Following calibration, the reference strains for the primary ligaments: ACL, LCL, MCL, and PCL were also variable (Fig. 4, Table 2), and did not exhibit clear trends for specific structures or teams, or knees. The reference strain for the PCL was more variable than that of the ACL in both DU02 and OKS003 models (Fig. 4).

3.2.2 Passive Flexion. In passive flexion, the trend in AP translation of the tibia relative to the femur followed the experimental behavior observed in the DU02 and OKS003 models for all teams (Fig. 5). Team HSS showed a larger anterior offset relative to the experimental reference and the results of others. While most teams (ABI, CC, DU, WSU) matched the VrVl trend observed in the experiment, some offsets from the experimental VrVl pose throughout flexion were noted. The results for IE were more variable. While four teams matched the increasing internal rotation experimental data of the DU02 knee, there was an offset in external tibial rotation for Teams ABI and HSS. The OKS003 experimental IE behavior was unconventional [25] and exhibited a small external rotation with flexion that few teams matched even after calibration and the Team HSS model failed to capture. RMSE between the model and experiment were quantified; smaller RMSE were noted for VrVl and AP, than for IE (compare calibrated results in Table 3), but differences between teams were also present.

Lastly, the impact of calibration was quantified by comparing the uncalibrated and calibrated model prediction for AP translation and IE and VrVl rotation during passive flexion (Fig. 5). For both knees and across teams, knee kinematics after calibration more closely followed the experimental data (compare dashed and solid lines in Fig. 5). For both knees, errors in deeper flexion (near 90 deg) were larger than near full extension. When comparing the RMSE, variability was observed between teams and between knees (Table 3). AP and VrVl RMSE decreased for all teams and both knee

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

Table 1 Calibration process details across the teams

Group	Knee	Number of design variables	Number of target points	Design variable type	Target data loads	Target DOF per target point	DOF in displacement control	Flexion angles calibrated (deg)	DOF in force control	Optimization cost function	Optimization constraints	Optimization penalty function	Optimization stopping criteria	Optimization algorithm type	Optimization algorithm
University of Auckland (ABI)	DU02	4	Time points from fitted data at 0 and 90	Ligament reference strain	Load at each time point from fitted data at 0 and 90	1	FE	0, 90	AP, IE, VrVl	Average of normalized RMSE of experimental to simulation targets	Reference strain: ACL 0.80–0.95, PCL 1.05– 1.20, MCL 0.89–1.04, LCL 0.93–1.00	Penalty if a simulation terminated before completion	Set amount of execution time	Global	TNC, L-BFGS-B & Powell (results with Powell)
	OKS003	4	Time points from fitted data at 0 and 90	Ligament reference strain	Load at each time point from fitted data at 0 and 90	1	FE	0, 90	AP, IE, VrVl	Average of normalized RMSE of experimental to simulation targets	Reference strain: ACL 0.75–0.95, PCL 1.05–1.25, MCL 0.9–1.05, LCL 0.85–1.05	Penalty if a simulation terminated before completion	Set amount of execution time	Global	TNC, L-BFGS-B & Powell (results with TNC)
Cleveland Clinic (CC)	DU02	4	44	Ligament reference strain	Selected from resampled data	1	FE	0	VrVl, IE, ML, AP, SI	RMSE of experimental to simulation targets	None	None	Change in prestrain less than 0.001	Local	Unconstrained minimization
	OKS003	4	32	Ligament reference strain	Selected from resampled data	1	FE	0	VrVl, IE, ML, AP, SI	RMSE of experimental to simulation targets	None	None	Change in prestrain less than 0.001	Local	Unconstrained minimization
University of Denver (DU)	DU02	26	24	Ligament stiffness and reference strain	Maximum and minimum loads	2	FE	10, 40, 60, 90	VrVl, IE, ML, AP, SI	RMSE of experimental to simulation targets	$\pm 10\%$ of the initial guess of the design variables	Penalty if sum of each ligament load across all target trials > 10N	Change in cost function and design variables $< 0.5\%$	Local	Nelder-Meade Simplex
	OKS003	26	24	Ligament stiffness and reference strain	Maximum and minimum loads	2	FE	0, 30, 60, 90	VrVl, IE, ML, AP, SI	RMSE of experimental to simulation targets	$\pm 10\%$ of the initial guess of the design variables	Penalty if sum of each ligament load across all target trials > 10N	Change in cost function and design variables $< 0.5\%$	Global	Simulated annealing

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

Table 1 (continued)

Group	Knee	Number of design variables	Number of target points	Design variable type	Target data loads	Target DOF per target point	DOF in displacement control	Flexion angles calibrated (deg)	DOF in force control	Optimization cost function	Optimization constraints	Optimization penalty function	Optimization stopping criteria	Optimization algorithm type	Optimization algorithm
Hospital for Special Surgery (HSS)	DU02	8	8	Ligament reference strain	Maximum and minimum inflection points	1	FE	0, 30, 90	VrVl, IE, ML, AP, SI	Fourth power of experimental and simulation target error	±10% of the initial guess of the design variables	None	Cost function <0.8	Local	Generalized reduced gradient
	OKS003	8	8	Ligament reference strain	Maximum and minimum inflection points	1	FE	0, 30, 90	VrVl, IE, ML, AP, SI	Fourth power of experimental and simulation target error	±10% of the initial guess of the design variables	None	Cost function <0.8	Local	Generalized reduced gradient
Washington State (WSU)	DU02	22	10	Ligament reference strain	Maximum and minimum loads	5	FE	0, 90	VrVl, IE, ML, AP, SI	Squared error of weighted experimental to simulation targets	None	At least one fiber in each ligament carries > 0.1 N	Change in objective function $< 1 \times 10^{-4}$	Local	Sequential least squares programing (SLSQP)
	OKS003	22	12	Ligament reference strain	Maximum and minimum loads	5	FE	0, 90	VrVl, IE, ML, AP, SI	Squared error of weighted experimental to simulation targets	None	At least one fiber in each ligament carries > 0.1 N	Change in objective function $< 1 \times 10^{-4}$	Local	Sequential least squares programing (SLSQP)

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

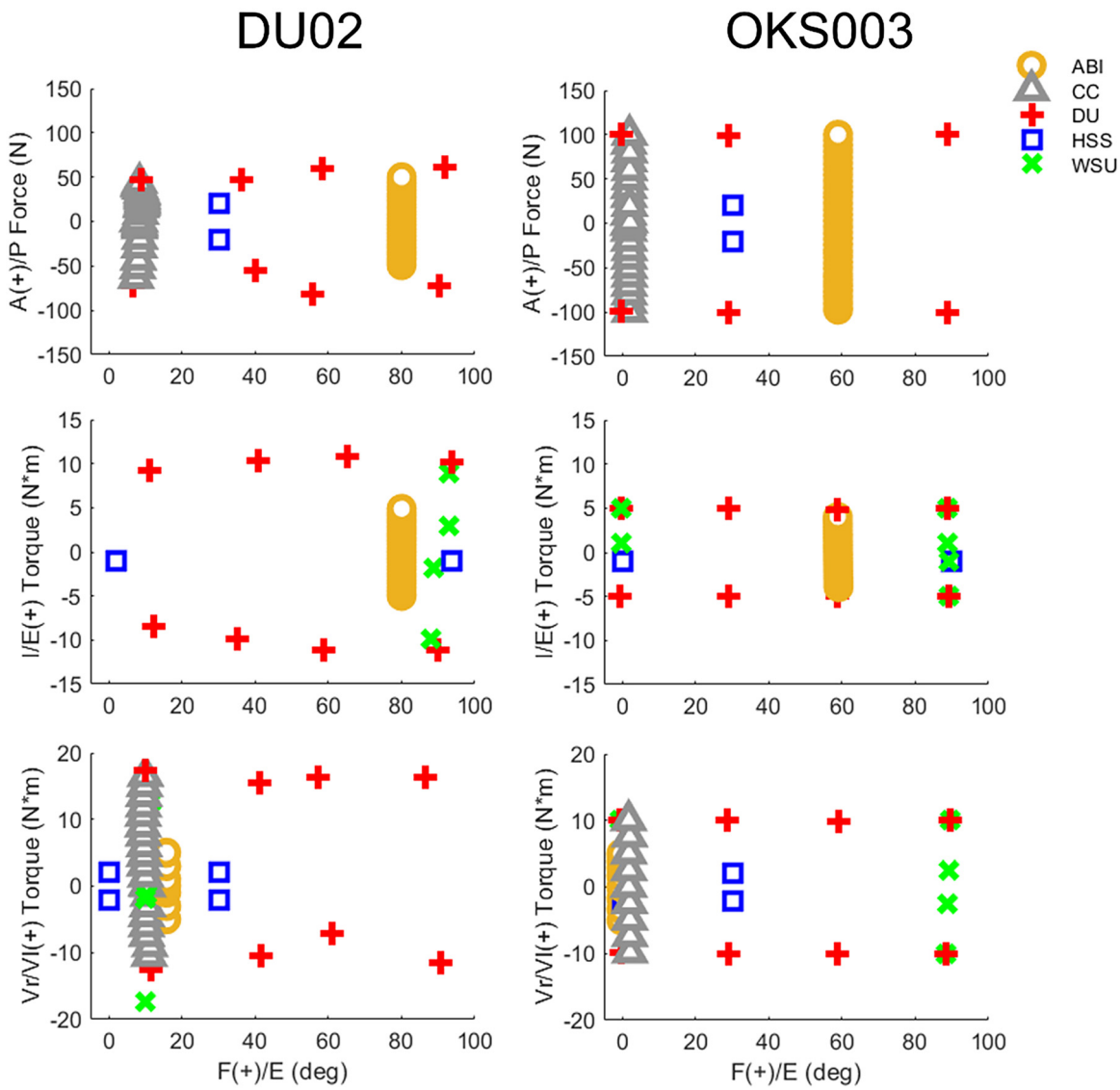

Fig. 3 Calibration targets derived from experimental knee laxity data and selected by each team for model calibration

models, with one exception in each knee model. The calibrated model results for 4 out of 5 teams had smaller errors in IE rotation for DU02 and OKS003 than the uncalibrated models (Table 3).

4 Discussion

Data sharing and reproducibility is a fundamental tenet in the responsible allocation of resources for impact on the field. By considering workflows for five teams, the KneeHub project built upon previous work to develop knee models and calibrate these models to experimental data. The range of calibration strategies and decisions were then compared to one another and documented, and their effects on knee kinematic predictions were quantified.

The kinematic behavior of the knee is influenced by the articular geometry and the constraints provided by the ligaments. In the current model calibration phase, teams performed calibration of the previously developed models to specified experimental data, which included laxity measurements of the knees under various knee flexion angles and loads. Teams used these measurements to calibrate the behavior of modeled ligamentous structures to replicate subject-specific kinematic-kinetic relationships. Teams were not given specific instructions on the calibration process to capture the full range of modeling methodologies across teams. Although all teams applied an optimization process to perform model calibration, notable differences in optimization strategies, calibration targets, design variables, cost functions, and stopping criteria demonstrated the range of approaches and strategies to calibrate computational models.

In the prior model development phase [6], while similar approaches were used to segment and develop the anatomical geometries, differences were observed in the modeling tools selected and the representation of key structures (e.g., ligaments), including modeling simplifications (1D linear representations versus 3D structures). Further, differences in the location of ligament attachments were noted between teams, which resulted from a combination of model decision-making and uncertainty in the interpretation of the imaging data, which presumably carried forward to influence this model calibration phase.

When representing structures, modelers will typically weigh accuracy and computational time as calibration via optimization requires a large number of simulations. For example, all teams represented bone as rigid bodies, and all teams except Team CC represented cartilage with rigid body approximations to improve computational efficiency with negligible loss in accuracy [16,33]. For the four primary knee ligament structures, team approaches were more widely distributed and ranged in strategy from 3D fiber-reinforced composite representations of the entire structure to a series of multiple 1D nonlinear springs representing bundles of the ligaments. While 3D ligament representations are thought to better represent the mechanical behavior of the tissue itself, as well as the

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

Table 2 Final calibrated ligament parameters across the teams. Note: units for stiffness vary between N/mm, N/strain or MPa (volumetric) depending on the model. Values of the reference strain of less than 1 correspond to initially slack, and greater than 1 represent initially taught.

Group	Knee	Representation	ACL			PCL			MCL			LCL		
University of Auckland (ABI)	DU02	4 3D continuum ligaments		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)
			ACL	0.846	123	PCL	1.416	168	MCL	0.972	224	LCL	1.202	280
	OKS003	4 3D continuum ligaments		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)
			ACL	0.750	123	PCL	1.250	168	MCL	1.050	224	LCL	0.923	280
Cleveland Clinic (CC)	DU02	4 3D continuum ligaments		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)
			ACL	1.016	225	PCL	1.0	283	MCL	1.073	74	LCL	1.015	47
	OKS003	4 3D continuum ligaments		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)		Ref Strain	Stiffness (MPs)
			ACL	1.081	225	PCL	0.926	283	MCL	0.984	74	LCL	1.062	47
University of Denver (DU)	DU02	30 1D point to point fibers		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)
			ACLam	0.9862	68.876	PCLal	0.8704	37.438	MCLa	1.0297	169.390	LCL	0.9506	131.490
			ACLpl	0.9350	59.671	PCLpm	0.8120	69.008	MCLm	1.0300	169.390			
									MCLp	1.0230	169.390			
	OKS003	30 1D point to point fibers		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)
			ACLam	1.1299	56.149	PCLal	0.9512	95.315	MCLa	0.9096	74.876	LCL	0.9740	162.190
			ACLpl	0.9674	103.310	PCLpm	0.9747	77.151	MCLm	0.8272	74.876			
									MCLp	0.8136	74.876			
Hospital for Special Surgery (HSS)	DU02	14 1D point to point fibers		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)
			ACLam	0.971	25.4	PCLal	1.140	57.0	MCLprox	0.920	80.0	LCL	0.920	59.0
			ACLpl	0.971	28.8	PCLpm	1.140	120.0	MCLdist	0.920	80.0			
	OKS003	14 1D point to point fibers		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)		Ref Strain	Stiffness (N/mm)
			ACLam	0.81	25.4	PCLal	0.561	57.0	MCLprox	0.875	80.0	LCL	1.013	59.0
			ACLpl	0.81	28.8	PCLpm	0.561	120.0	MCLdist	0.875	80.0			
Washington State (WSU)	DU02	32 1D point to point fibers		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)
			ACLam0	0.954	2120	PCLal0	0.957	5625	MCLprox0	0.842	1375	LCL0	0.842	2000
			ACLam1	0.843	2120	PCLal1	1.043	5625	MCLprox1	0.842	1375	LCL1	0.858	2000
			ACLpl0	0.842	2880	PCLpm0	0.999	3375	MCLdist0	1.018	1375			
			ACLpl1	0.951	2880	PCLpm1	1.050	3375	MCLdist1	0.963	1375			
	OKS003	32 1D point to point fibers		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)		Ref Strain	Stiffness (N/strain)
			ACLam0	0.969	2120	PCLal0	1.053	5625	MCLprox0	1.014	1375	LCL0	0.915	2000
			ACLam1	1.053	2120	PCLal1	1.053	5625	MCLprox1	0.844	1375	LCL1	0.905	2000
			ACLpl0	1.052	2880	PCLpm0	1.053	3375	MCLdist0	0.976	1375			
			ACLpl1	1.051	2880	PCLpm1	1.053	3375	MCLdist1	1.048	1375			

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

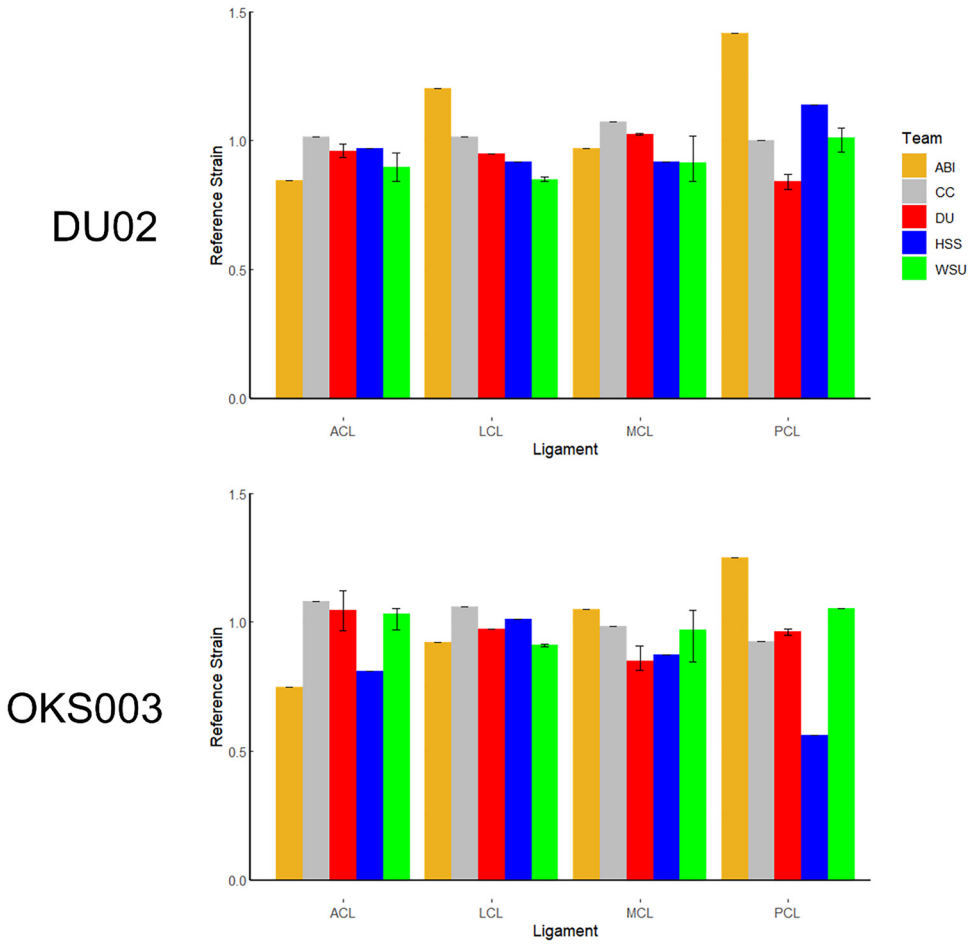

Fig. 4 Calibrated reference strains for primary knee ligaments. Graphs represent mean reference strain, with error bars representing the maximum and minimum reference strains for teams using multiple fiber bundles for the same ligament. Values of the reference strain of less than 1 correspond to initially slack, and greater than 1 represent initially taught.

wrapping and contact with other structures [16,34–36], they are more computationally intensive [35], and simplified models have been shown to efficiently replicate their stabilizing role within the knee [17,24,35,37]. Given the impact on computational resources, decisions related to the representation of structures also impacted the downstream optimization strategy, including the number of parameters, selection of the optimization algorithm, number of trials, and convergence criteria.

The most notable differences between teams during model calibration were observed in the selection of the calibration targets, and these decisions are directly linked to the "art" of modeling. To minimize computational costs and calibrate the key parameters, all teams attempted to use the fewest number of experimental data targets to sufficiently represent the mechanics of the knees (Fig. 3). Some teams selected specific flexion angles and degrees-of-freedom to calibrate specific structures. Teams with 3D ligament structures (Team ABI and CC) used more data points at a specified flexion angle and, given computational considerations, selected targets to emphasize the behavior of the ligament. In contrast, teams using multiple 1D ligament representations tended to use data from multiple flexion angles to accurately capture the loading of different bundles as the knee flexed. In these cases, the calibration targets were individual experimental values at various flexion angles; some teams selected intermediate values (HSS and WSU), while others chose extents (DU) for use in tuning the ligament properties. Taken together, the chosen calibration targets captured modelers' thoughts on the aspects and regions that were most important to describe the kinematic-kinetic behavior. The differences in chosen calibration targets impacted model performance during calibration and will likely further affect the model benchmarking and model reuse phases, emphasizing the need for detailed reporting and guidance on best practices in the community.

All teams were able to successfully develop calibrated models based on their selected calibration targets. While teams used optimization methods to tune parameters and develop viable models, differences were noted between teams in which parameters were tuned versus held constant. Variability was observed in the values for stiffness and reference strain for the four primary knee ligaments (Table 2). The lack of clear trends in the resultant ligament reference strains (Fig. 4) is likely due to differences in the selected stiffness or modulus and the specific attachment sites in each team's model. Several studies have quantified how changes in stiffness, reference strain, and attachment site impacted the laxity response, kinematics, and contact mechanics [38,39]. Conversely, in this study, when the ligament attachment sites differed among models [6], the observed variability in reference strains was to be expected as it may compensate for variations in attachment sites, and thus overall ligament length. Considering the range of parameter values for the teams, it will be interesting to assess the sensitivity of the model outputs to changes in the ligament parameters, but there are challenges in capturing the parameter interaction effects and accounting for differences in the representations and models in a large-scale way. Noting teams' models have different architectures inherited from model development, a future study could isolate and investigate the impact of the various calibration strategies by predicting model performance within a single team's model

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

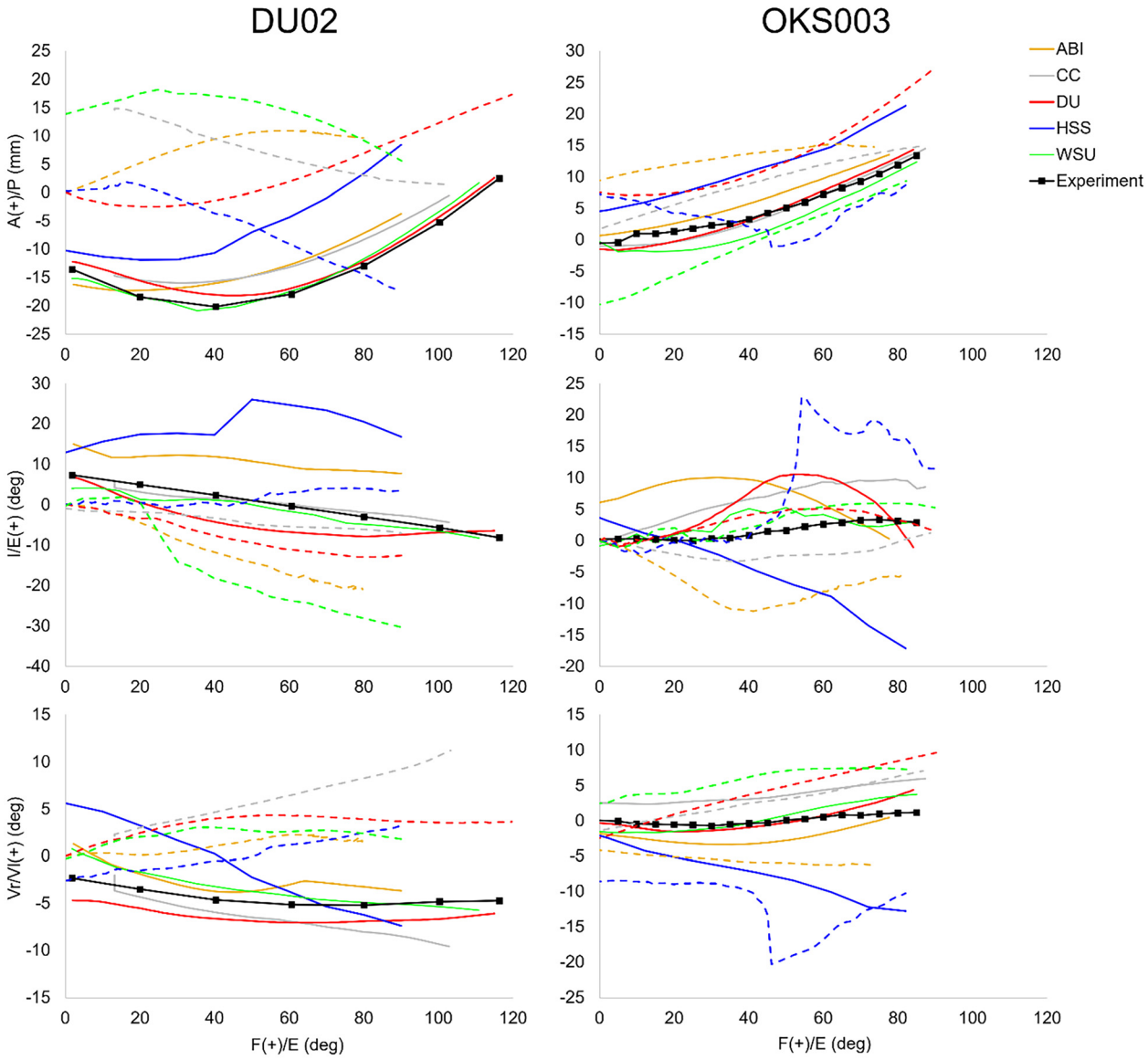

Fig. 5 Passive flexion kinematics as a function of knee flexion angle. The black line represents the original experimental kinematics, solid lines show the predicted kinematics after calibration, and dashed lines represent predicted kinematics from the end of the model development phase from Rooks et al. [6]. Kinematics from each team have been converted to a consistent coordinate system matching the experimental data for the given model.

architecture. The open-source sharing of our models and workflows supports this and other such investigations in the future. A key takeaway is that despite the observed variability in the parameters, the resulting models were largely able to describe the overall AP, IE, and VrVl kinematic trends for the knees as they are observed experimentally (Table 3, Fig. 5).

Calibration plays an important role in the development of models with subject specificity. Differences between subject anatomy, mechanical properties, and loading conditions are inherently present. In this work, standardized imaging and cadaveric experimental data were provided for model development. The calibration process resulted in better kinematic agreement and smaller RMSE between uncalibrated and calibrated models with few exceptions, but it also demonstrated the limits of fit due to the "art" of calibration, which dictates the resolution one can achieve for knee-specificity depending on the calibration approach. Substantial differences were observed in the experimental behavior of the selected specimens; in particular, OKS003 did not exhibit the screw home mechanism [40–43] that is commonly observed and was present in DU02. Decisions on approach and calibration targets influenced model ability to capture the experimental behavior and differences between subjects. In addition, across all teams, kinematic errors increased with deeper knee flexion; these errors, along with the noted subject-specific differences, suggest that models may need enhanced fidelity (such as additional anatomic structures) or greater emphasis on these regions in the calibration process. Accurate modeling of knee kinematic behavior over the entire flexion range is important and deep flexion is present in certain activities of daily living and challenges both healthy subjects and those with joint pathology or replacement.

Beyond data sharing, a further consideration in reproducibility is access to computational tools. In the workflow, some teams used open-source software (e.g., Febio, 3D Slicer) [44,45], while other teams used commercial software platforms (e.g., Abaqus, MSC Adams, Mimics, Simpleware). Similarly, a range of open-source, commercial, and homegrown optimization codes were used. In the model development phase, the end product was a mesh developed from segmentations, and the differences resulting from the tools used were small. In the model calibration phase, differences in the functionality of the analysis platforms were more substantial and

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

Table 3 RMSE relative to the experimental kinematics during passive flexion. Comparison of uncalibrated, from Rooks et al. [6], and calibrated team predictions to experimental data for the DU02 and OKS003 knees.

		AP RMSE (mm)		IE RMSE (deg)		VrVl RMSE (deg)	
Model	Team	Uncalibrated	Calibrated	Uncalibrated	Calibrated	Uncalibrated	Calibrated
DU02	ABI	27.05	4.11	14.76	9.25	5.95	1.67
	CC	27.03	4.09	1.54	1.12	10.08	1.67
	DU	18.46	1.71	9.03	5.92	8.37	1.99
	HSS	14.07	11.40	9.35	20.14	5.00	4.24
	WSU	33.70	0.54	19.47	2.04	7.19	1.27
OKS003	ABI	9.48	2.07	21.95	7.27	5.53	2.44
	CC	4.58	1.05	4.78	5.01	2.62	3.43
	DU	7.26	1.16	17.58	5.35	4.04	0.69
	HSS	4.45	6.76	13.91	7.73	12.68	7.56
	WSU	5.84	2.55	20.29	2.07	5.82	1.15

may have had a greater impact on modeling decisions and the kinematic results.

When looking forward, particularly when considering clinical applications, the ability of the model to describe knee kinematics and kinetics, and appropriately distribute load amongst the structures, will be further vetted in the model benchmarking and model reuse phases, which include more complex combined loading conditions and removal of structures (e.g., ACL deficient). In model reproducibility and drawing on the ASME V&V 40 guidelines [46], the model results should be evaluated with respect to the context of use and accuracy thresholds, which are defined as part of the model reuse phase.

Lastly, the conventions used to describe kinematics were not consistent between teams and experiments, which made direct comparisons challenging and highlighted the need for unified, or even standardized, data and modeling conventions in the biomechanics community.

The experiments utilized different kinematic conventions, namely, clinical GS for the DU02 data, and cylindrical GS for the OKS003 data. Each team converted the experimental data to match the coordinate system of their chosen modeling strategy [6]; with teams ABI, DU, and HSS using clinical GS as the primary coordinate system, and CC and WSU using cylindrical GS. Further, teams used different sign conventions for positive DOF (e.g., internal versus external rotation) and for whether the knee was a right (DU02) or left (OKS003). Future kinematic comparisons could be facilitated by reporting experimental data and model predictions as 4×4 homogenous transformation matrices between bones for each time point with a unified representation of bone coordinate systems. This approach would allow end users to change the coordinate systems to match their preferences while alleviating confusion.

In closing, reproducibility remains a concern in knee modeling and simulation, especially as these tools are becoming more widely adopted for scientific discovery and patient care. In this way, the findings of the KneeHub project go beyond knee modeling and have broader implications for the reproducibility and credibility of simulation-based research and medicine. This project and other studies [30,47,48] show that open-source approaches allow for more detailed descriptions and sharing of models and enable other community members to evaluate model approaches and accuracy in the context of their specific research questions and use cases. Calibration improved model kinematic tracking during passive knee flexion, even though teams selected varying calibration targets resulting in different calibrated ligament parameters. Complete project documentation and results are publicly available [26]. These collaborative efforts are key to understanding the modeling processes and parameters that influence reproducibility, establishing best practices for model calibration and data sharing, and advancing the practice of credible modeling in the biomechanics community.

Author Contribution Statement

TA, PL, and KS outlined the document, wrote the first full draft, and acquired input from coauthors. All coauthors added content in relevant sections, reviewed and edited the draft and confirmed their agreement with the contents of the document.

Acknowledgment

The views, opinions and/or findings contained in this document are those of the authors and do not necessarily reflect the views of the funding agencies.

Funding Data

- Contributions of the authors were primarily supported by the National Institute of Biomedical Imaging and Bioengineering, National Institutes of Health (NIH) (Grant No. R01EB024573; Funder ID: 10.13039/100000070).
- Efforts by AE were also supported in part by NIH R01GM104139, and KS by NIH R01EB015497 (Funder ID: 10.13039/100000002).

Conflict of Interest

Coauthors did not report any conflicts of interest with the content of this article.

Data Availability Statement

The data and information that support the findings of this article are freely available (link available in footnote[2]).

References

[1] Collins, F. S., and Tabak, L. A., 2014, "NIH Plans to Enhance Reproducibility," Nature, **505**(7485), pp. 612–613.

[2] Peng, R. D., Dominici, F., and Zeger, S. L., 2006, "Reproducible Epidemiologic Research," Am. J. Epidemiol., **163**(9), pp. 783–789.

[3] Open Science Collaboration, 2015, "Psychology. Estimating the Reproducibility of Psychological Science," Science, **349**(6251), p. aac4716.

[4] National Academies of Sciences, Engineering, and Medicine; Policy and Global Affairs; Committee on Science, Engineering, Medicine, and Public Policy; Board on Research Data and Information; Division on Engineering and Physical Sciences; Committee on Appli, and S. S. C. on R. and R. in S., 2019, "Understanding Reproducibility and Replicability," *Reproducibility and Replicability in Science*, National Academies Press (U.S.), Washington, DC.

[5] Erdemir, A., Besier, T. F., Halloran, J. P., Imhauser, C. W., Laz, P. J., Morrison, T. M., and Shelburne, K. B., 2019, "Deciphering the 'Art' in Modeling and Simulation of the Knee Joint: Overall Strategy," ASME J. Biomech. Eng., **141**(7), p. 071002.

[6] Rooks, N. B., Schneider, M. T. Y., Erdemir, A., Halloran, J. P., Laz, P. J., Shelburne, K. B., Hume, D. R., Imhauser, C. W., Zaylor, W., Elmasry, S., Schwartz, A., Chokhandre, S. K., Abdollahi Nohouji, N., and Besier, T. F., 2021,

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023

"Deciphering the 'Art' in Modeling and Simulation of the Knee Joint: Variations in Model Development," ASME J. Biomech. Eng., **143**(6), p. 061002.
[7] Mulugeta, L., Drach, A., Erdemir, A., Hunt, C. A., Horner, M., Ku, J. P., Myers, J. G., Vadigepalli, R., and Lytton, W. W., 2018, "Credibility, Replicability, and Reproducibility in Simulation for Biomedicine and Clinical Applications in Neuroscience," Front. Neuroinform., **12**(April), pp. 1–16.
[8] Peters, A. E., Akhtar, R., Comerford, E. J., and Bates, K. T., 2018, "Tissue Material Properties and Computational Modelling of the Human Tibiofemoral Joint: A Critical Review," PeerJ, **2018**(1), pp. 1–48.
[9] Taylor, M., and Prendergast, P. J., 2015, "Four Decades of Finite Element Analysis of Orthopaedic Devices: Where Are We Now and What Are the Opportunities?," J. Biomech., **48**(5), pp. 767–778.
[10] Erdemir, A., Guess, T. M., Halloran, J., Tadepalli, S. C., and Morrison, T. M., 2012, "Considerations for Reporting Finite Element Analysis Studies in Biomechanics," J. Biomech., **45**(4), pp. 625–633.
[11] Balci, O., 2012, "A Life Cycle for Modeling and Simulation," Simulation, **88**(7), pp. 870–883.
[12] Henak, C. R., Anderson, A. E., and Weiss, J. A., 2013, "Subject-Specific Analysis of Joint Contact Mechanics: Application to the Study of Osteoarthritis and Surgical Planning," ASME J. Biomech. Eng., **135**(2), p. 021003.
[13] Guan, S., Gray, H. A., Keynejad, F., and Pandy, M. G., 2016, "Mobile Biplane X-Ray Imaging System for Measuring 3D Dynamic Joint Motion During Overground Gait," IEEE Trans. Med. Imaging, **35**(1), pp. 326–336.
[14] Liukkonen, M. K., Mononen, M. E., Tanska, P., Saarakkala, S., Nieminen, M. T., and Korhonen, R. K., 2017, "Application of a Semi-Automatic Cartilage Segmentation Method for Biomechanical Modeling of the Knee Joint," Comput. Methods Biomech. Biomed. Eng., **20**(13), pp. 1453–1463.
[15] Maletsky, L., Shalhoub, S., Fitzwater, F., Eboch, W., Dickinson, M., Akhbari, B., and Louie, E., 2015, "In Vitro Experimental Testing of the Human Knee: A Concise Review," J. Knee Surg., **29**(2), pp. 138–148.
[16] Trad, Z., Barkaoui, A., Chafra, M., and Tavares, J. M. R. S., 2018, *FEM Analysis of the Human Knee Joint*, Springer, Cham.
[17] Blankevoort, L., and Huiskes, R., 1996, "Validation of a Three-Dimentional Model of the Knee," J. Biomech., **29**(7), pp. 955–961.
[18] Andreassen, T. E., Hamilton, L. D., Hume, D., Higinbotham, S. E., Behnam, Y., Clary, C., and Shelburne, K. B., 2021, "Apparatus for In Vivo Laxity Assessment Using High-Speed Stereo Radiography," ASME J. Med. Devices, **15**(4), p. 041004.
[19] Kupper, J. C., Westover, L., Frayne, R., and Ronsky, J. L., 2016, "Application of a Novel Measure of In Vivo Knee Joint Laxity," ASME J. Biomech. Eng., **138**(10), p. 104501.
[20] Markolf, K. L., Mensch, J. S., and Amstutz, H. C., 1976, "Stiffness and Laxity of the Knee - The Contributions of the Supporting Structures," J. Bone Jt. Surg., **58**(5), pp. 583–594.
[21] Emery, M., Moffroid, M., Boerman, J., Fleming, B., Howe, J., and Pope, M., 1989, "Reliability of Force/Displacement Measures in a Clinical Device Designed to Measure Ligamentous Laxity at the Knee," J. Orthop. Sports Phys. Ther., **10**(11), pp. 441–447.
[22] Daniel, D. M., Stone, M. L., Sachs, R., and Malcom, L., 1985, "Instrumented Measurement of Anterior Knee Laxity in Patients With Acute Anterior Cruciate Ligament Disruption," Am. J. Sports Med., **13**(6), pp. 401–407.
[23] Blankevoort, L., Huiskes, R., and de Lange, A., 1988, "The Envelope of Passive Knee Joint Motion," J. Biomech., **21**(9), pp. 705–720.
[24] Harris, M. D., Cyr, A. J., Ali, A. A., Fitzpatrick, C. K., Rullkoetter, P. J., Maletsky, L. P., and Shelburne, K. B., 2016, "A Combined Experimental and Computational Approach to Subject-Specific Analysis of Knee Joint Laxity," ASME J. Biomech. Eng., **138**(8), p. 081004.
[25] Chokhandre, S., Schwartz, A., Klonowski, E., Landis, B., and Erdemir, A., 2023, "Open Knee(s): a Free and Open Source Library of Specimen-Specific Models and Related Digital Assets for Finite Element Analysis of the Knee Joint," Ann. Biomed. Eng., **51**(1), pp. 10–23.
[26] SimTK, 2018, "Reproducibility in Simulation-Based Prediction of Natural Knee Mechanics: Project Home," SimTK, accessed Dec. 1, 2018, https://simtk.org/projects/kneehub
[27] University of Denver, 2018, "Natural Knee Data," Center for Orthopaedic Biomechanics, University of Denver, Denver, CO, accessed Dec. 1, 2018, https://digitalcommons.du.edu/natural_knee_data/
[28] SimTK, 2018, "Open Knee(s): Virtual Biomechanical Representations of the Knee Joint: Specifications," accessed Jan. 12, 2018, https://simtk.org/plugins/moinmoin/openknee/Specifications/ExperimentationAnatomicalImaging
[29] SimTK, 2018, "Open Knee(s): Virtual Biomechanical Representations of the Knee Joint: Specifications," accessed Jan. 12, 2018, https://simtk.org/plugins/moinmoin/openknee/Specifications/ExperimentationJointMechanics
[30] Halloran, J. P., Abdollahi Nohouji, N., Hafez, M. A., Besier, T. F., Chokhandre, S. K., Elmasry, S., Hume, D. R., Imhauser, C. W., Rooks, N. B., Schneider, M. T. Y., Schwartz, A., Shelburne, K. B., Zaylor, W., and Erdemir, A., 2023, "Assessment of Reporting Practices and Reproducibility Potential of a Cohort of Published Studies in Computational Knee Biomechanics," J. Orthop. Res., **41**(2), pp. 325–334.
[31] Rooks, N. B., Schneider, M. T. Y., Erdemir, A., Halloran, J. P., Laz, P. J., Shelburne, K. B., Hume, D. R., Imhauser, C. W., Zaylor, W., Elmasry, S., Schwartz, A., Chokhandre, S. K., Abdollahi Nohouji, N., and Besier, T. F., 2021, "A Method to Compare Heterogeneous Types of Bone and Cartilage Meshes," ASME J. Biomech. Eng., **143**(11), p. 111002.
[32] Grood, E. S., and Suntay, W. J., 1983, "A Joint Coordinate System for the Clinical Description of Three-Dimensional Motions: Application to the Knee," ASME J. Biomech. Eng., **105**(2), pp. 136–144.
[33] Halloran, J. P., Petrella, A. J., and Rullkoetter, P. J., 2005, "Explicit Finite Element Modeling of Total Knee Replacement Mechanics," J. Biomech., **38**(2), pp. 323–331.
[34] Galbusera, F., Freutel, M., Durselen, L., D'Aiuto, M., Croce, D., Villa, T., Sansone, V., and Innocenti, B., 2014, "Material Models and Properties in the Finite Element Analysis of Knee Ligaments: A Literature Review," Front. Bioeng. Biotechnol., **2**(November), pp. 1–11.
[35] Orozco, G. A., Tanska, P., Mononen, M. E., Halonen, K. S., and Korhonen, R. K., 2018, "The Effect of Constitutive Representations and Structural Constituents of Ligaments on Knee Joint Mechanics," Sci. Rep., **8**(1), pp. 1–15.
[36] Naghibi Beidokhti, H., Janssen, D., van de Groes, S., Hazrati, J., Van den Boogaard, T., and Verdonschot, N., 2017, "The Influence of Ligament Modelling Strategies on the Predictive Capability of Finite Element Models of the Human Knee Joint," J. Biomech., **65**, pp. 1–11.
[37] Weiss, J. A., Gardiner, J. C., Ellis, B. J., Lujan, T. J., and Phatak, N. S., 2005, "Three-Dimensional Finite Element Modeling of Ligaments: Technical Aspects," Med. Eng. Phys., **27**(10), pp. 845–861.
[38] Baldwin, M. A., Laz, P. J., Stowe, J. Q., and Rullkoetter, P. J., 2009, "Efficient Probabilistic Representation of Tibiofemoral Soft Tissue Constraint," Comput. Methods Biomech. Biomed. Eng., **12**(6), pp. 651–659.
[39] Rooks, N. B., Besier, T. F., and Schneider, M. T. Y., 2022, "A Parameter Sensitivity Analysis on Multiple Finite Element Knee Joint Models," Front. Bioeng. Biotechnol., **10**(May), pp. 1–11.
[40] Hamilton, L. D., Andreassen, T. E., Myers, C., Shelburne, K. B., Clary, C., and Rullkoetter, P. J., 2022, "Supine Leg Press as an Alternative to Standing Lunge in High-Speed Stereo Radiography," J. Biomech., **138**(April), p. 111118.
[41] Kim, H. Y., Kim, K. J., Yang, D. S., Jeung, S. W., Choi, H. G., and Choy, W. S., 2015, "Screw-Home Movement of the Tibiofemoral Joint During Normal Gait: Three-Dimensional Analysis," CiOS Clin. Orthop. Surg., **7**(3), pp. 303–309.
[42] Asano, T., Akagi, M., Tanaka, K., Tamura, J., and Nakamura, T., 2001, "In Vivo Three-Dimensional Knee Kinematics Using a Biplanar Image-Matching Technique," Clin. Orthop. Relat. Res., **388**, pp. 157–166.
[43] Hallén, L. G., and Lindahl, O., 1966, "The 'Screw-Home' Movement in the Knee-Joint," Acta Orthop. Scand., **37**(1), pp. 97–106.
[44] Maas, S. A., Ellis, B. J., Ateshian, G. A., and Weiss, J. A., 2012, "FEBio: Finite Elements for Biomechanics," ASME J. Biomech. Eng., **134**(1), p. 011005.
[45] Fedorov, A., Beichel, R., Kalpathy-Cramer, J., Finet, J., Fillion-Robin, J.-C., Pujol, S., Bauer, C., Jennings, D., Fennessy, F., Sonka, M., Buatti, J., Aylward, S., Miller, J. V., Pieper, S., and Kikinis, R., 2012, "3D Slicer as an Image Computing Platform for the Quantitative Imaging Network," Magn. Reson. Imag., **30**(9), pp. 1323–1341.
[46] ASME, 2018, *Assessing Credibility of Computational Modeling Through Verification and Validation: Application to Medical Devices*, ASME, New York.
[47] Begley, C. G., and Ioannidis, J. P. A., 2015, "Reproducibility in Science: Improving the Standard for Basic and Preclinical Research," Circ. Res., **116**(1), pp. 116–126.
[48] Imhauser, C. W., Baumann, A. P., Liu, X. (., Bischoff, J. E., Verdonschot, N., Fregly, B. J., Elmasry, S. S., Abdollahi, N. N., Hume, D. R., Rooks, N. B., Schneider, M. T., Zaylor, W., Besier, T. F., Halloran, J. P., Shelburne, K. B., and Erdemir, A., 2023, "Reproducibility in Modeling and Simulation of the Knee: Academic, Industry, and Regulatory Perspectives," J. Orthop. Res., epub.

Downloaded from http://asmedigitalcollection.asme.org/biomechanical/article-pdf/145/12/121008/7054924/bio_145_12_121008.pdf by University of Denver, Thor Andreassen on 31 October 2023